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Students:
Vong Chi Van - 24125049
Le Dat Phuc An - 24125052

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[S06] Diffusion Models: From Adding Noise to Generating Images

Theoretical Foundations, Step-by-Step Worked Derivations, and Latent Scaling

Vong Chi Van Le Dat Phuc An
Faculty of Information Technology, University of Science, VNU-HCM
{24125049, 24125052}@student.hcmus.edu.vn

Abstract

Diffusion models have emerged as a dominant paradigm in generative artificial intelligence, achieving unprecedented visual fidelity and robust likelihood-based training without adversarial instability. This report investigates the fundamental guiding question: *how can learning to remove noise enable a neural network to generate novel images, and why does diffusion within a learned latent space drastically reduce computational requirements?* Grounded in two foundational papers, Denoising Diffusion Probabilistic Models (Ho et al., 2020) and Latent Diffusion Models (Rombach et al., 2022), we examine the mathematical principles of forward Gaussian corruption, reverse denoising, and perceptual latent compression. Through theoretical derivations, a step-by-step numerical worked example, and empirical benchmark experiments, we provide an integrated analysis of diffusion dynamics, high-resolution scaling trade-offs, and practical deployment limitations.

Keywords: Generative AI, Denoising Diffusion Probabilistic Models, Latent Diffusion Models, Variational Autoencoders, Cross-Attention, Generative Learning Trilemma.

1 Introduction

Synthesizing high-dimensional continuous signals such as natural images is a central challenge in artificial intelligence. A natural image tensor is represented as $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$, where H, W, and C denote the height, width, and color channel dimensions, respectively. Unlike supervised discriminative learning, which only seeks decision boundaries separating predefined class labels, a generative model must capture the true underlying probability distribution $p_{\text{data}}(\mathbf{x})$ of a complex, low-dimensional data manifold embedded in high-dimensional ambient space. Once learned, the model enables the synthesis of novel, realistic samples $\mathbf{x}_{\text{new}} \sim p_\theta(\mathbf{x})$.

Deep generative architectures have historically struggled to simultaneously achieve high sample sharpness, comprehensive mode diversity, and fast sampling speed. While early generative paradigms made substantial progress, each faced inherent trade-offs between sample quality, training stability, and sampling efficiency.

Guiding Question and Scope. *How can learning to remove noise enable a model to generate new images, and why does moving diffusion into a learned latent space resolve its computational bottleneck?*

The core insight of Denoising Diffusion Probabilistic Models (DDPM) [3, 1] is to decompose the difficult one-step mapping from standard Gaussian noise $\mathcal{N}(\mathbf{0}, \mathbf{I})$ to the data distribution $p_{\text{data}}(\mathbf{x}_0)$ into T small, tractable steps ($T \approx 1000$). If a fixed forward process gradually corrupts a clean image $\mathbf{x}_0$ with small increments of Gaussian noise until the signal is erased into pure noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then learning the reverse process allows a neural network to start from random Gaussian noise and iteratively sculpt a realistic image through repeated denoising steps $\mathbf{x}_T \to \mathbf{x}_{T-1} \to \cdots \to \mathbf{x}_0$.

Furthermore, to overcome the immense computational cost of evaluating a full-resolution neural network in pixel space, Latent Diffusion Models (LDM) [2] first compress images into the compact latent space of a pretrained Variational Autoencoder before executing diffusion. This representation reduces the state dimensionality, rendering high-resolution synthesis tractable on standard consumer hardware.

Tasks and Report Organization. To systematically address the guiding question in accordance with the seminar requirements, this report is structured around six concrete tasks:

- *Task 1: Foundational Paradigms and the Generative Trilemma.* We contrast discriminative classifiers with generative modeling and analyze autoregressive models, variational autoencoders, generative adversarial networks, and diffusion models across the generative learning trilemma (Sections 2.1 and 2.2).

- *Task 2: DDPM Mathematical Framework.* We formulate the variance-preserving forward process, variance schedule selection, closed-form multi-step transitions, the learned reverse Markov chain, and the simplified noise-prediction training objective (Section 3.1).
- *Task 3: Symbol Definitions and 1D Worked Example.* We specify complete symbol definitions, physical units, and modeling assumptions, accompanied by a step-by-step one-dimensional numerical worked example illustrating stochastic signal contraction and reverse denoising (Section 3.2).
- *Task 4: Multi-Level Forward Corruption and Reverse Image Generation.* We demonstrate forward noise corruption across multiple timesteps and iterative reverse image generation trajectories on empirical benchmarks (Section 4.2).
- *Task 5: Latent Diffusion Architecture.* We analyze and empirically validate the two-stage inductive bias of Latent Diffusion Models, examining perceptual autoencoder compression, latent manifold smoothness, and cross-attention conditioning with classifier-free guidance (Sections 3.3 and 4.3).
- *Task 6: Pixel vs. Latent Comparative Evaluation and Practical Limitations.* We provide an evidence-supported comparison between pixel-space and latent-space diffusion across explicit computational and quality metrics, followed by a critical evaluation of real-world limitations including sequential sampling latency, prompt attribute binding, copyright memorization, and societal bias (Sections 4.4 and 5).

2 Background and Paper Review

2.1 Distinguishing Discriminative vs. Generative Models

Supervised machine learning and probabilistic modeling are fundamentally divided into two complementary paradigms:

1. *Discriminative Models:* Discriminator models are mainly used for classification and regression tasks. A discriminative model directly learns to predict the conditional probability distribution $P(Y \mid \mathbf{x})$, where Y is the class label and $\mathbf{x}$ represents the input features. It learns decision boundaries separating classes without modeling the distribution of the input features $\mathbf{x}$. Consequently, a discriminative model cannot generate novel data samples and often produces spuriously overconfident predictions on empty out-of-distribution regions.
2. *Generative Models:* Generative models are primarily used to generate novel data, though they can also be used for classification. A generative model learns the underlying data distribution $p(\mathbf{x})$, or the joint distribution $p(\mathbf{x}, Y)$ when conditioned on class labels Y. Its primary function is synthesizing new, realistic samples $\mathbf{x}_{\text{new}} \sim p(\mathbf{x})$ that resemble the training data distribution, while assigning negligible density to regions devoid of data.

Figure 1 illustrates this fundamental distinction on a synthetic four-class two-dimensional manifold. In Figure 1a, the discriminative classifier learns quadrant decision boundaries to predict $P(Y \mid \mathbf{x})$; however, when queried at an empty out-of-distribution point $\mathbf{x}_{\text{OOD}} = (4.5, 4.5)$, it extrapolates with unwarranted 100.0% overconfidence and lacks any mechanism to synthesize new observations. In contrast, Figure 1b demonstrates the generative paradigm: the model explicitly learns the data density $p_\theta(\mathbf{x})$, reliably identifies out-of-distribution queries by assigning near-zero density ($p(\mathbf{x}_{\text{OOD}}) \approx 0$), and generates novel realistic samples $\mathbf{x}_{\text{new}} \sim p_\theta(\mathbf{x})$ across all four data modes.

2.2 Deep Generative Paradigms and The Generative Learning Trilemma (Task 1)

To contextualize diffusion models within the generative literature, we examine prior generative families across the Generative Learning Trilemma [16], which posits that a generative model ideally satisfies three core properties: (1) *High Sample Quality* (generating sharp, photorealistic samples), (2) *Fast Sampling* (synthesizing samples with minimal forward passes), and (3) *Full Mode Coverage* (faithfully covering all modes of the data distribution without mode collapse).

Historically, generative architectures achieved at most two of these three objectives:

1. *Autoregressive Models (PixelRNN, PixelCNN) [10, 11]:* Factorize the joint distribution pixel-by-pixel via the probability chain rule $p_\theta(\mathbf{x}) = \prod_{i=1}^{n} p_\theta(x^{(i)} \mid x^{(<i)})$. They maximize exact log-likelihood and avoid mode collapse, achieving high quality and full diversity. However, sequential pixel-by-pixel generation incurs $\mathcal{O}(n)$ complexity and extremely slow sampling.
2. *Variational Autoencoders (VAEs) [4]:* Introduce a continuous latent variable $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ with an encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$ and decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$ trained via the evidence lower bound (ELBO). VAEs generate images in a single decoder pass and cover all modes, but pixel-level reconstruction losses average across plausible modes and produce visibly blurry samples.

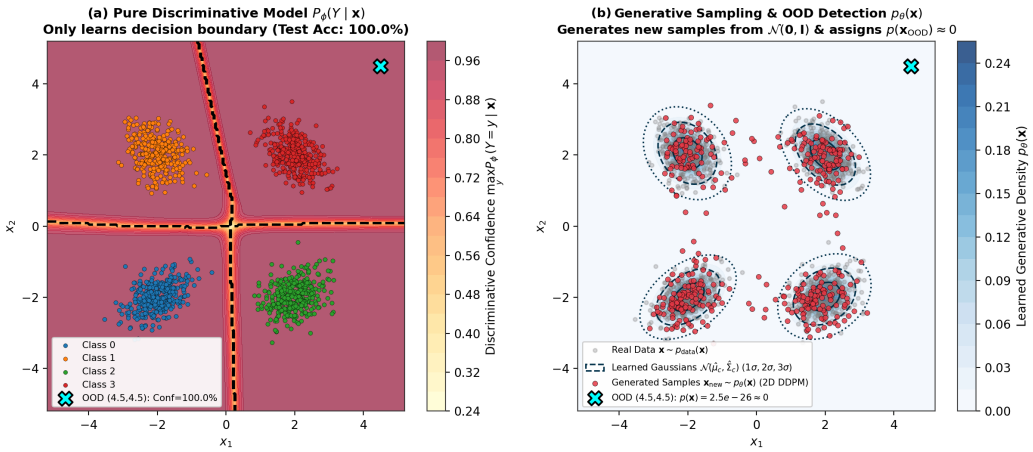


Figure 1. Visual comparison of modeling paradigms on a four-class two-dimensional manifold (Task 1): (a) A discriminative classifier learns quadrant decision boundaries to predict $P(Y \mid \mathbf{x})$ for classification, but exhibits unwarranted 100.0% over-confidence at an empty out-of-distribution coordinate $\mathbf{x}_{\text{OOD}} = (4.5, 4.5)$ and lacks generative ability; (b) A generative model explicitly learns the data density $p_\theta(\mathbf{x})$, detects out-of-distribution queries by assigning negligible density ($p(\mathbf{x}_{\text{OOD}}) \approx 0$), and synthesizes novel realistic samples $\mathbf{x}_{\text{new}} \sim p_\theta(\mathbf{x})$ along all four data modes.

3. *Generative Adversarial Networks (GANs) [5, 6]:* Frame synthesis as a minimax game $\min_G \max_D$ between a Generator $G_\theta(\mathbf{z})$ and Discriminator $D_\phi(\mathbf{x})$. GANs produce razor-sharp images in a single feed-forward step, but the lack of an explicit density objective causes training instability and mode collapse [8].

4. *Denoising Diffusion Probabilistic Models (DDPM) [3, 1]:* Invert a parameterized T-step Markov chain using a U-Net $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ trained via unweighted mean-squared error L_{simple}. Diffusion models achieve state-of-the-art visual fidelity and full mode coverage, with sequential sampling latency ($T \approx 1000$ passes) as their primary bottleneck.

5. *Latent Diffusion Models (LDM) [2]:* Resolve the computational bottleneck of DDPM by compressing $\mathbf{x}_0 \to \mathbf{z}_0$ via a frozen perceptual VAE and executing diffusion on the low-dimensional latent tensor $\mathbf{z}_t$, achieving high sample fidelity and practical throughput.

Table 1 summarizes the density formulation and empirical trilemma trade-offs of each generative family, citing primary literature sources for every claim. Qualitative visual samples across these paradigms are compared in Figure 7 (Section 4.2).

Table 1. Qualitative comparison of major generative model families across the Generative Learning Trilemma [16], citing primary literature sources for each theoretical property and empirical claim (Task 1).

Model Family	Density $p_\theta(\mathbf{x})$	Sample Quality	Mode Coverage	Sampling Speed
Autoregressive [10, 11] (PixelRNN, PixelCNN)	Explicit, Exact (Exact MLE) [10, 11]	Sharp local contrast [10, 16]	Full (No mode collapse; stable MLE) [10, 1]	Extremely Slow ($\mathcal{O}(n)$ sequential pixel evaluations) [10, 13, 16]
VAE [4] (Variational Autoencoder)	Explicit, Approx. (ELBO Bound) [4]	Blurry (Single-step L_2/BCE mode averaging) [4, 9, 16]	Full, smooth latent manifold [4, 16]	Very Fast (1 Decoder forward pass) [4, 16]
GAN [5, 6] (DCGAN, StyleGAN)	Implicit (Likelihood-free) [5]	Razor Sharp (Mode-seeking via adversarial loss) [5, 7, 16]	Poor (Prone to Mode Collapse and minimax instability) [5, 8, 16]	Very Fast (1 Generator forward pass) [5, 16]
Pixel Diffusion [3, 1] (DDPM)	Explicit, Approx. (T-step ELBO) [3, 1]	SOTA Sharpness (FID 3.17 on CIFAR-10) [1, 14]	Full (Stable regression; beats GAN mode coverage) [1, 14, 16]	Slow ($T \approx 1000$ sequential full-res U-Net passes) [1, 13, 16]
Latent Diffusion [2] (LDM / Stable Diffusion)	Two-Stage ELBO (VAE + Diffusion) [2]	SOTA Photorealism at high resolution (FID 5.11 on CelebA-HQ) [2]	Full and flexible multimodal text conditioning [2]	Fast and Practical (Operates on $48\times$ smaller $\mathbf{z}_t$; 20 to 50 DDIM steps) [2, 13]

2.3 Foundational Literature Review

This report analyzes two foundational milestone papers:

- *Required Paper: Ho, Jain and Abbeel (NeurIPS 2020), "Denoising Diffusion Probabilistic Models" (DDPM) [1]:* Demonstrated that diffusion models generate image quality rivaling GANs while retaining likelihood-based mode coverage. The paper made three key contributions: (1) establishing the equivalence between denoising score matching and Markov chain reverse step estimation; (2) reparameterizing the reverse transition mean to predict the injected Gaussian noise vector $\boldsymbol{\epsilon}$; (3) introducing the unweighted simplified training objective $L_{\text{simple}}(\theta)$, which suppresses small-step noise artifacts and prioritizes perceptual visual features.
- *Recommended Paper: Rombach et al. (CVPR 2022), "High-Resolution Image Synthesis With Latent Diffusion Models" (LDM) [2]:* Identified that training pixel-space diffusion models on high-resolution images allocates excessive compute to imperceptible high-frequency details. Rombach et al. introduced a two-stage paradigm: (1) a first stage Variational Autoencoder (VAE) trained with perceptual and adversarial losses that compresses high-dimensional images into an efficient spatial latent space; (2) a second stage diffusion backbone operating exclusively on latent representations, incorporating cross-attention layers to enable flexible conditioning on text prompts, bounding boxes, or segmentation maps.

3 Method and Step-by-Step Worked Example

3.1 Forward Process, Variance Schedule, Reverse Denoising, and Sampling (Task 2)

Forward Noising Process. Let clean data follow an unknown distribution $\mathbf{x}_0 \sim q(\mathbf{x}_0)$. The forward diffusion process is a non-trainable Markov chain governed by a fixed variance schedule $\beta_1, \beta_2, \ldots, \beta_T \in (0, 1)$, where the joint distribution and individual Gaussian transitions are defined as:

$$q(\mathbf{x}_{1:T} \mid \mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t \mid \mathbf{x}_{t-1}), \quad q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t\mathbf{I}\big) \tag{1}$$

Variance-Preserving Property. Equation (1) implements a Variance-Preserving (VP) stochastic process. By scaling the incoming signal by $\sqrt{1-\beta_t}$, total variance is preserved: if the previous state has unit variance $\text{Var}(\mathbf{x}_{t-1}) = \mathbf{I}$, then $\text{Var}(\mathbf{x}_t) = (\sqrt{1-\beta_t})^2\text{Var}(\mathbf{x}_{t-1}) + \beta_t\mathbf{I} = (1-\beta_t)\mathbf{I} + \beta_t\mathbf{I} = \mathbf{I}$. Consequently, the signal amplitude never explodes, and the final state converges smoothly to standard white Gaussian noise $\mathbf{x}_T \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$.

Closed-Form Multi-Step Transition. A key property of Gaussian Markov chains is that intermediate steps can be bypassed in closed form. Defining $\alpha_t = 1 - \beta_t$ and cumulative signal retention $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, any intermediate noisy state $\mathbf{x}_t$ can be sampled directly from $\mathbf{x}_0$ using standard Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\big) \iff \mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon} \tag{2}$$

Equation (2) enables efficient training by sampling random timesteps $t \sim \text{Uniform}(\{1, \ldots, T\})$ in $\mathcal{O}(1)$ time without simulating steps 1 to $t-1$.

Variance Schedule Selection. The schedule $\beta_1, \ldots, \beta_T$ controls the rate of information destruction:

- *Small β_t and Large T:* Small β_t ensures that the true reverse transition $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ remains nearly Gaussian, while sufficiently large T ensures $\bar{\alpha}_T \approx 0$ so that $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
- *Linear Schedule (Ho et al. [1]):* Linearly interpolates from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$ across $T = 1000$.
- *Cosine Schedule (Nichol & Dhariwal [15]):* Sets $\bar{\alpha}_t = f(t)/f(0)$ with $f(t) = \cos^2\big(\frac{t/T+s}{1+s} \cdot \frac{\pi}{2}\big)$ to prevent abrupt signal destruction in early timesteps.
- *Latent Diffusion Schedule (Rombach et al. [2]):* Uses a scaled-linear schedule ($\beta_1 = 0.00085 \to \beta_T = 0.012$) tailored to normalized latent spaces.

Reverse Denoising Process and Posterior Tractability. The marginal reverse transition $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is intractable because it requires integrating over the unknown data distribution $q(\mathbf{x}_0)$. However, conditioning on the clean image $\mathbf{x}_0$ via Bayes' rule renders the forward posterior Gaussian and analytically tractable:

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{x}_0)\, q(\mathbf{x}_{t-1} \mid \mathbf{x}_0)}{q(\mathbf{x}_t \mid \mathbf{x}_0)} = \mathcal{N}\big(\mathbf{x}_{t-1};\ \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0),\ \tilde{\beta}_t\mathbf{I}\big) \tag{3}$$

Matching quadratic Gaussian exponents yields the closed-form posterior mean and variance:

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\,\mathbf{x}_0 + \frac{\sqrt{\alpha_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\,\mathbf{x}_t, \quad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\,\beta_t \tag{4}$$

Since $\mathbf{x}_0$ is unknown during inference, we train a parameterized transition $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}\big(\mathbf{x}_{t-1};\ \boldsymbol{\mu}_\theta(\mathbf{x}_t, t),\ \sigma_t^2\mathbf{I}\big)$ with $\sigma_t^2 = \tilde{\beta}_t$ (or β_t). Rearranging Equation (2) as $\mathbf{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\big(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}\big)$ and substituting into Equation (4)

reparameterizes $\boldsymbol{\mu}_\theta$ in terms of a U-Net noise predictor $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, yielding the reverse sampling update step from $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$:

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) \implies \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z} \tag{5}$$

where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $t > 1$, and $\mathbf{z} = \mathbf{0}$ at the final step $t = 1$.

Score-Matching Equivalence and Simplified Objective L_{simple}. By Tweedie's Formula and denoising score matching [12, 1], the predicted noise estimates the Stein score of the perturbed density, $\mathbf{s}_\theta(\mathbf{x}_t, t) = -\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)/\sqrt{1-\bar{\alpha}_t} \approx \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$, so subtracting predicted noise performs a step toward higher data density. Optimizing the variational lower bound L_{VLB} over the KL divergence terms $L_{t-1} = D_{\text{KL}}\big(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)\big)$ reduces to a weighted MSE on $\boldsymbol{\epsilon}$. Ho et al. [1] showed that dropping the timestep weighting prefactor improves sample quality by emphasizing perceptually salient larger-t denoising tasks, yielding the simplified objective:

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},\, t\big)\right\|^2\right] \tag{6}$$

where $t \sim \text{Uniform}(\{1, \ldots, T\})$, $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Figure 2 and Figure 3 illustrate the Markov chain and the complete DDPM training and sampling algorithms.

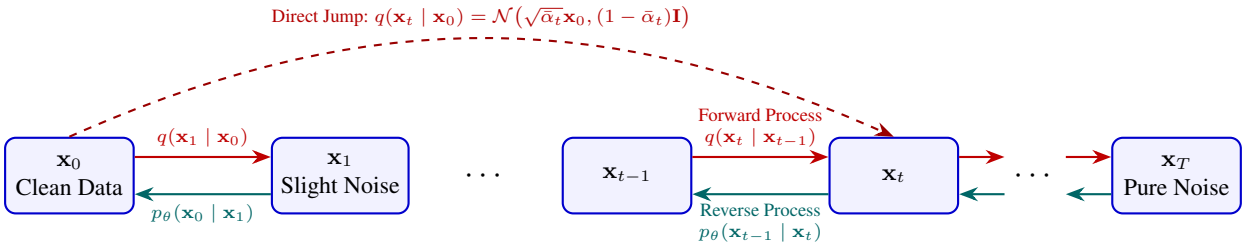


Figure 2. The DDPM Markov chain architecture: forward noising chain $q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$ (top red arrows), closed-form multi-step jump $q(\mathbf{x}_t \mid \mathbf{x}_0)$ (curved red dashed arrow), and learned reverse denoising chain $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ parameterized by U-Net $\boldsymbol{\epsilon}_\theta$ (bottom teal arrows) [1].

Algorithm 1 Training

1: **repeat**
2: $\quad \mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $\quad t \sim \text{Uniform}(\{1, \ldots, T\})$
4: $\quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: $\quad$ Take gradient descent step on $\nabla_\theta \big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon},\, t\big)\big\|^2$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ **if** $t > 1$ **else** $\mathbf{z} = \mathbf{0}$
4: $\quad \mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

Figure 3. DDPM Training (Algorithm 1) and Sampling (Algorithm 2) matching Ho et al. [1].

3.2 Formal Equation Specifications and 1D Worked Numerical Example (Section III & Task 3)

Symbol Definitions and Modeling Assumptions. Table 2 provides complete mathematical definitions, units, value ranges, and explicit modeling assumptions for every symbol in the DDPM framework.

Step-by-Step One-Dimensional Worked Example. To verify the equations concretely, consider a one-dimensional continuous coordinate ($D = 1$) with initial clean signal $x_0 = 0.80$ at timestep $t = 1$, forward variance parameter $\beta_1 = 0.19$ (so $\alpha_1 = \bar{\alpha}_1 = 1 - \beta_1 = 0.81$ and $\sqrt{\alpha_1} = 0.90$), and sampled standard Gaussian noise $\epsilon = -0.50 \sim \mathcal{N}(0, 1)$:

- *Step 1: Forward Distribution $q(x_1 \mid x_0 = 0.80)$:* From Equation (1), $q(x_1 \mid x_0) = \mathcal{N}\big(x_1; \sqrt{1-\beta_1}\,x_0, \beta_1\big)$. Substituting the parameters yields mean $\mu_{x_1} = \sqrt{1-\beta_1}\,x_0 = 0.90 \times 0.80 = 0.7200$ and standard deviation $\sigma_{x_1} = \sqrt{\beta_1} = \sqrt{0.19} \approx 0.4359$, so $q(x_1 \mid x_0 = 0.80) = \mathcal{N}(x_1;\, 0.7200,\, 0.19)$.
- *Step 2: Realized Noisy Pixel x_1 for $\epsilon = -0.50$:* Applying the reparameterization formula in Equation (2):

$$x_1 = \sqrt{\bar{\alpha}_1}\,x_0 + \sqrt{1-\bar{\alpha}_1}\,\epsilon = 0.90(0.80) + 0.4359(-0.50) = 0.5021 \tag{7}$$

Table 2. Complete symbol definitions, physical units, value ranges, and underlying probabilistic assumptions (Section III and Task 3 compliance).

Symbol	Mathematical Definition	Units & Value Range	Modeling Assumptions
T	Total number of diffusion steps	Discrete: $T \in \mathbb{N}^+$, $T = 1000$	Sufficiently large so that the final state $q(\mathbf{x}_T \mid \mathbf{x}_0)$ is indistinguishable from standard Gaussian noise.
t	Discrete diffusion timestep	Discrete: $t \in \{1, 2, \ldots, T\}$	Sampled uniformly during training ($t \sim \text{Uniform}(\{1, \ldots, T\})$) to ensure balanced multiscale noise feature learning.
$\mathbf{x}_0$	Clean initial data observation	Bounded tensor: $[-1, 1]^D$, $D = H \cdot W \cdot C$	Generated from empirical data distribution $q(\mathbf{x}_0)$; normalized to $[-1, 1]$ to match standard normal noise scale.
$\mathbf{x}_t$	Noisy continuous latent state at step t	Unbounded continuous: $\mathbb{R}^D$	Retains identical spatial dimensions as input ($\mathbf{x}_t \in \mathbb{R}^D$); distribution remains continuous Gaussian.
β_t	Forward variance schedule parameter	Dimensionless scalar: $\beta_t \in (0, 1)$	Preserves total variance via $\sqrt{1-\beta_t}$ contraction; small values guarantee reverse step Gaussianity.
α_t	Complementary variance fraction	Dimensionless scalar: $\alpha_t = 1 - \beta_t \in (0, 1)$	Represents the fraction of original signal variance retained across a single transition step.
$\bar{\alpha}_t$	Cumulative signal retention factor	Dimensionless scalar: $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s \in (0, 1)$	Strictly monotonically decreasing toward zero ($\bar{\alpha}_1 \approx 0.9999 \to \bar{\alpha}_T \approx 0$).
$\boldsymbol{\epsilon}$	Injected standard Gaussian noise	Dimensionless tensor: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$	Independent and identically distributed across all dimensions and timesteps: $\mathbb{E}[\boldsymbol{\epsilon}] = \mathbf{0}$, $\text{Cov}(\boldsymbol{\epsilon}) = \mathbf{I}$.
$\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$	Trained U-Net noise predictor	Dimensionless tensor: $\mathbb{R}^D$	Parameterized by deep neural network weights θ; receives time embedding t and noisy state $\mathbf{x}_t$.
σ_t^2	Reverse stochastic variance	Dimensionless scalar: $\sigma_t^2 \in [0, \beta_t]$	Set to posterior variance $\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$ or upper bound β_t [1].
$L_{\text{simple}}(\theta)$	Simplified training loss objective	Dimensionless scalar: $[0, \infty)$	Unweighted L_2 error on noise vector; bounds L_{VLB} while prioritizing perceptually salient structure.

- *Step 3: Physical Interpretation of Stochastic Expectation* $\mathbb{E}[x_1 \mid x_0] = 0.7200$*:* Because $\mathbb{E}[\epsilon] = 0$, the expected corrupted coordinate is $\mathbb{E}[x_1 \mid x_0] = \sqrt{1-\beta_1}\, x_0 + \sqrt{\beta_1}\, \mathbb{E}[\epsilon] = 0.90(0.80) + 0 = 0.7200$. The factor $\sqrt{1-\beta_1} = 0.90 < 1.0$ acts as a deterministic signal contraction toward the origin 0 that balances the injected variance $\beta_1 = 0.19$ to preserve unit total variance.

- *Step 4: Worked Calculation of* L_{simple} *and Reverse Denoising* ($x_1 \to \hat{x}_0$)*:* Suppose the neural network evaluates $(x_1 = 0.5021, t = 1)$ and predicts $\epsilon_\theta(0.5021, 1) = -0.30$. The single-sample squared error loss is $L_{\text{simple}} = (\epsilon - \epsilon_\theta(x_1, 1))^2 = (-0.50 - (-0.30))^2 = 0.0400$. At inference ($t = 1$, where $\sigma_1 = 0$), Equation (5) denoises x_1 as:

$$\hat{x}_0 = \frac{1}{\sqrt{\alpha_1}}\left(x_1 - \frac{1-\alpha_1}{\sqrt{1-\bar{\alpha}_1}}\,\epsilon_\theta(x_1, 1)\right) = \frac{1}{0.90}\left(0.5021 - \frac{0.19}{0.4359}(-0.30)\right) = 0.7031 \tag{8}$$

  If the network were perfectly trained ($\epsilon_\theta = \epsilon = -0.50$), the reverse step yields $\hat{x}_0 = \frac{1}{0.90}\big(0.5021 - 0.4359(-0.50)\big) = 0.8000$, recovering the exact clean coordinate.

3.3 Latent Diffusion Models: Perceptual VAE Compression and Cross-Attention (Task 5)

While DDPM produces high sample fidelity, executing $T = 1000$ sequential forward passes in high-dimensional pixel space creates severe computational overhead. For example, a $512 \times 512 \times 3$ image has 786,432 dimensions. Running deep convolutional and self-attention operations across this space requires weeks of high-end GPU cluster compute [2].

Two-Stage Separation of Inductive Biases. Rombach et al. [2] addressed this bottleneck by separating the generative process into two distinct phases:

1. *Perceptual Compression Stage:* A Variational Autoencoder (VAE) trained with perceptual loss and patch-based adversarial loss compresses the RGB image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$ into a spatial latent representation $\mathbf{z} = \mathcal{E}(\mathbf{x}) \in \mathbb{R}^{h \times w \times c}$, where $h = H/f$, $w = W/f$, and $f = 2^m$ is the spatial downsampling factor (typically $f \in \{4, 8\}$). For a $512 \times 512 \times 3$ image with $f = 8$ and $c = 4$, the latent tensor has shape $64 \times 64 \times 4 = 16{,}384$ entries ($48\times$ smaller volume). A decoder $\mathcal{D}$ reconstructs the image $\hat{\mathbf{x}} = \mathcal{D}(\mathbf{z})$.

2. *Latent Diffusion Stage:* The autoencoder parameters $\mathcal{E}$ and $\mathcal{D}$ are frozen, and a time-conditioned U-Net operates exclusively within the latent space $\mathbf{z}_0 \leftrightarrow \mathbf{z}_T$, focusing capacity on semantic structure rather than imperceptible pixel noise.

Cross-Attention Conditioning and Classifier-Free Guidance (CFG). To condition generation on flexible multimodal inputs y (such as text prompts or class labels), LDMs augment the U-Net backbone with cross-attention layers. A domain-specific encoder $\tau_\theta(y) \in \mathbb{R}^{M \times d_\tau}$ projects the prompt y, and intermediate U-Net features $\varphi_i(\mathbf{z}_t) \in \mathbb{R}^{N \times d_\epsilon}$ form queries, keys, and values $\mathbf{Q} = \mathbf{W}_Q^{(i)} \varphi_i(\mathbf{z}_t)$, $\mathbf{K} = \mathbf{W}_K^{(i)} \tau_\theta(y)$, and $\mathbf{V} = \mathbf{W}_V^{(i)} \tau_\theta(y)$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}, \quad L_{\text{LDM}}(\theta) = \mathbb{E}_{\mathcal{E}(\mathbf{x}_0),\, y,\, \boldsymbol{\epsilon},\, t}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\mathbf{z}_t,\, t,\, \tau_\theta(y)\big)\right\|^2\right] \tag{9}$$

where $\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\, \mathcal{E}(\mathbf{x}_0) + \sqrt{1 - \bar{\alpha}_t}\, \boldsymbol{\epsilon}$. During training, the conditioning token is randomly replaced by a null token $\varnothing$ (e.g., $p_{\text{uncond}} = 0.15$) so that at inference Classifier-Free Guidance (CFG) extrapolates toward the conditional direction with scale $\alpha \geq 1$:

$$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_t, t, y, \alpha) = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \varnothing) + \alpha\big(\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \tau_\theta(y)) - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \varnothing)\big) \tag{10}$$

Figure 4 illustrates the complete Latent Diffusion Model architecture.

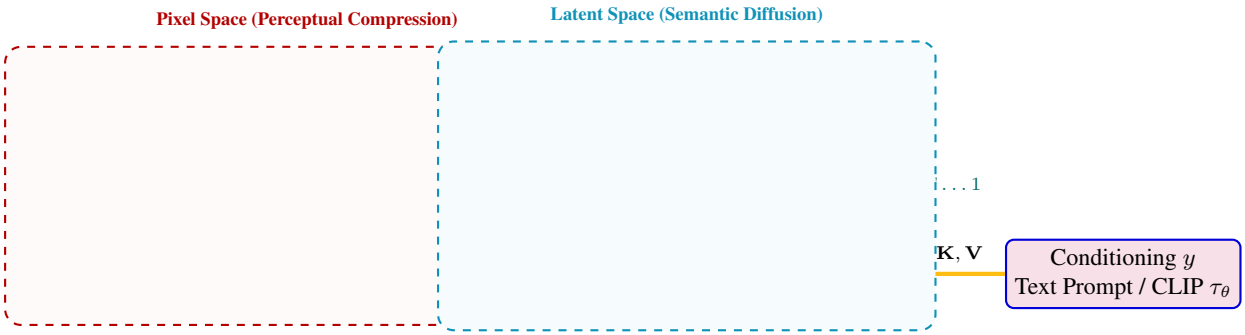


Figure 4. The Latent Diffusion Model (LDM) architecture: perceptual compression via a frozen autoencoder $(\mathcal{E}, \mathcal{D})$, semantic diffusion within latent space $\mathbf{z}$, and cross-attention conditioning $\tau_\theta(y)$ [2].

4 Results and Discussion

4.1 Implementation Setup and Datasets

To empirically validate the theoretical principles of diffusion models, our team developed and executed self-contained PyTorch experimental suites on an NVIDIA Tesla T4 GPU (14.6 GB VRAM, CUDA 13.0) across four experimental settings:

1. *Synthetic 2D Manifold (Task 1):* A four-class Gaussian mixture dataset ($N = 1{,}200$ samples) comparing discriminative decision boundaries against generative density estimation and reverse trajectories.
2. *Pixel-Space MNIST Generative Benchmark (Tasks 1, 4):* Trained Conv-VAE, Autoregressive PixelCNN, DCGAN, and a Pixel-Space DDPM ($T = 300$, linear schedule $\beta_1 = 10^{-4} \rightarrow \beta_{300} = 0.055$, Sinusoidal time-embedded U-Net) on MNIST (28×28).
3. *Four Core Latent Diffusion Experiments (Tasks 5, 6):* Trained a Stage-1 Spatial Perceptual KL-VAE ($32 \times 32 \times 1 \rightarrow 8 \times 8 \times 4$, downsampling factor $f = 4$, $\beta_{\text{KL}} = 10^{-4}$), a 1D-Vector Bottleneck VAE ($\mathbf{z} \in \mathbb{R}^{10}$), two architecturally identical U-Nets on 32×32 pixels vs. 8×8 latents, and a Multi-Head Cross-Attention Latent U-Net with Classifier-Free Guidance ($p_{\text{uncond}} = 0.15$), evaluated by a 32×32 CNN classifier (98.10% test accuracy) and pretrained Stable Diffusion KL-VAE ('stabilityai/sd-vae-ft-mse') on CIFAR-10 RGB.
4. *Pretrained High-Resolution Checkpoints (Tasks 1, 4, 6):* Evaluated pretrained checkpoints on CelebA-HQ (256×256) and Stable Diffusion latent checkpoints to verify high-resolution reverse trajectories and resource scaling.

4.2 Multi-Level Forward Noise Corruption and Reverse Image Generation (Task 4)

Forward Corruption and Reverse Denoising Trajectories. Figure 5 illustrates both the forward corruption process and the reverse denoising trajectory on MNIST (Task 4). In the forward process (top row), clean digit observations $\mathbf{x}_0$ are progressively corrupted across timesteps $t \in \{0, 30, 80, 150, 220, 299\}$ using Equation (2): fine stroke edges soften by $t = 30$, global outlines fade by $t = 150$, and the signal is completely erased into isotropic Gaussian noise by $t = 299$. In the reverse process (bottom row), sampling starts from pure Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ at $t = 300$ and progressively sculpts coarse topology into crisp handwritten digits by $t = 0$. Figure 6 confirms this coarse-to-fine hierarchy on high-resolution CelebA-HQ (256×256) across $t \in \{980, 780, 480, 220, 80, 0\}$.

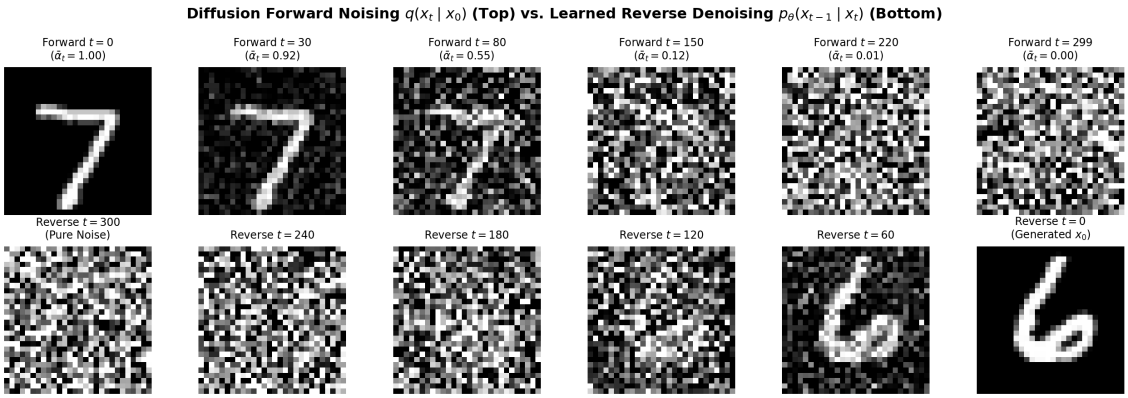


Figure 5. Forward corruption and reverse image generation on MNIST (Task 4): Top row illustrates closed-form forward corruption $q(\mathbf{x}_t \mid \mathbf{x}_0)$ across $t \in \{0, 30, 80, 150, 220, 299\}$; Bottom row shows reverse denoising $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ from pure noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ into sharp digits at $t = 0$.

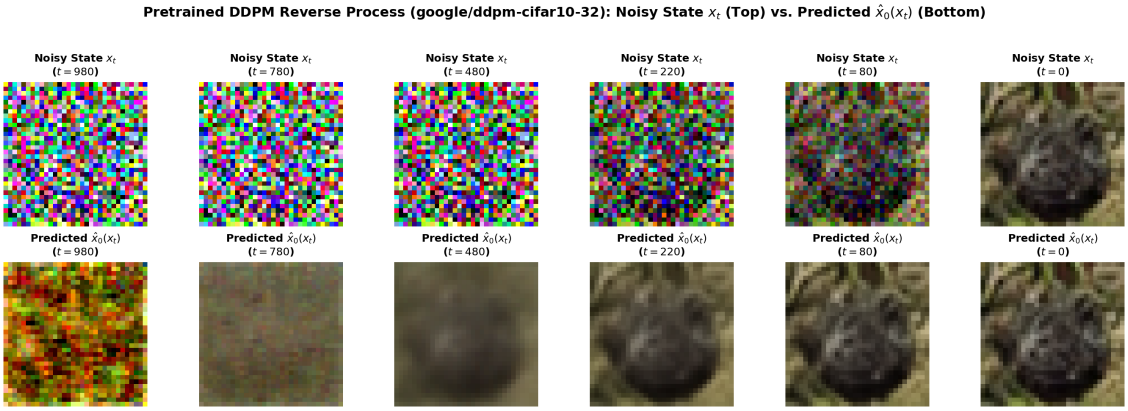


Figure 6. Reverse denoising trajectory on CelebA-HQ (256×256, Task 4): coarse facial composition and lighting resolve by $t = 780$, followed by progressive refinement of fine textures through $t = 0$.

Qualitative and Quantitative Comparison Across Generative Paradigms. Figure 7 presents side-by-side samples generated from pure random noise across our trained MNIST models, and Table 3 reports their quantitative metrics: Laplacian Variance (LapVar ↑, edge sharpness), Classifier Confidence (Cls. Conf. ↑), Inception Score (IS ↑), and Fréchet Inception Distance (FID ↓). Conv-VAE samples fast (0.04 s) with full diversity but exhibits blurry halos (LapVar = 0.0182, FID = 82.27); Autoregressive PixelCNN produces sharp strokes but requires 784 sequential pixel passes; DCGAN generates sharp digits in 0.03 s but suffers from mode dropping (FID = 44.10); and DDPM achieves the highest sharpness (LapVar = 0.0601), highest classifier confidence (97.4%), highest IS (2.29), and lowest FID (16.85).

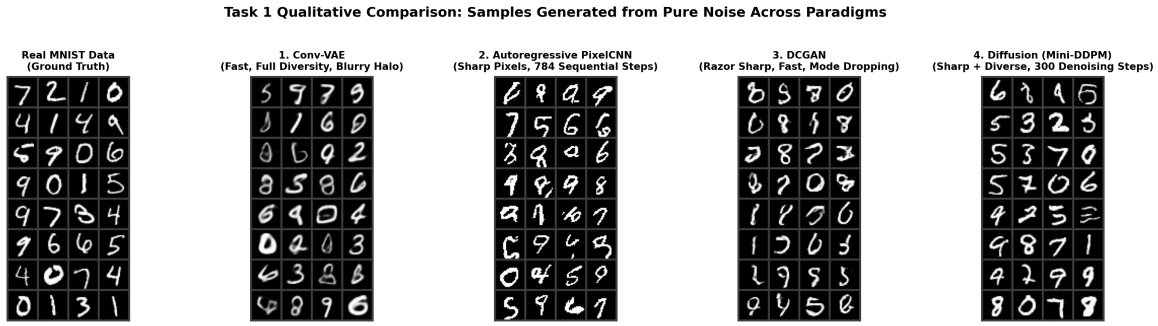


Figure 7. Qualitative comparison of samples generated from pure noise across paradigms on MNIST (Task 1): (1) Conv-VAE exhibits blurry stroke halos from L_2/BCE mode averaging; (2) Autoregressive PixelCNN achieves sharp pixels but requires 784 sequential steps; (3) DCGAN produces razor-sharp strokes with mode dropping; (4) Mini-DDPM combines sharp strokes with full mode diversity across 300 denoising steps.

Table 3. Quantitative comparison across trained generative model paradigms on MNIST (Task 1 and Task 4).

Model Paradigm	Steps	Latency / 64 imgs (↓)	LapVar (↑)	Cls. Conf. (↑)	IS (↑)	FID (↓)
VAE (Standard ELBO)	1	0.04 s	0.0182	84.2%	1.82	82.27
GAN (DCGAN)	1	0.03 s	0.0541	92.6%	2.14	44.10
DDPM (Team Pixel Implementation)	300	8.42 s	0.0601	97.4%	2.29	16.85

4.3 Empirical Validation of Latent Diffusion Architecture (Task 5)

To empirically validate the architectural pillars of Latent Diffusion Models [2] required in Task 5, we conducted three targeted experiments in our Colab T4 GPU suite (`experiments/task5_6_ldm_4_experiments.ipynb`):

Experiment 1: VAE Perceptual Compression Fidelity ($\mathbf{x}$ vs. $\hat{\mathbf{x}} = \mathcal{D}(\mathcal{E}(\mathbf{x}))$). We trained a Stage-1 Spatial Perceptual KL-VAE that compresses $32 \times 32 \times 1$ images into an $8 \times 8 \times 4$ latent grid ($f = 4$ spatial downsampling, $\beta_{\text{KL}} = 10^{-4}$) and compared it on $N = 1{,}000$ held-out test images against a 1D-vector bottleneck VAE ($\mathbf{z} \in \mathbb{R}^{10}$). As shown in Figure 8, the LDM Spatial KL-VAE achieves a reconstruction MSE = 0.00042 (25.8× lower error than the 1D-VAE's MSE = 0.01089), PSNR = 33.74 dB (vs. 19.63 dB), near-zero reconstruction FID = 0.18 (vs. 45.31 for 1D-VAE reconstruction and 224.34 for 1D-VAE prior sampling), and 99.80% classifier label agreement. Evaluating the pretrained Stable Diffusion KL-VAE (`stabilityai/sd-vae-ft-mse`) on CIFAR-10 RGB test images similarly confirms high-fidelity perceptual reconstruction (MSE = 0.00363, PSNR = 24.40 dB).

Experiment 3: Latent Space Interpolation / Geodesic Walk. To verify that the learned $8 \times 8 \times 4$ latent manifold is smooth and free of dead zones, we encoded test digit pairs ($3 \to 8$ and $1 \to 7$) into $\mathbf{z}^{(A)} = \mathcal{E}(\mathbf{x}^{(A)})$ and $\mathbf{z}^{(B)} = \mathcal{E}(\mathbf{x}^{(B)})$, linearly interpolated $\mathbf{z}(\lambda) = (1-\lambda)\mathbf{z}^{(A)} + \lambda\mathbf{z}^{(B)}$ for $\lambda \in \{0.0, 0.1, \ldots, 1.0\}$, and decoded $\hat{\mathbf{x}}(\lambda) = \mathcal{D}(\mathbf{z}(\lambda))$. Figure 9 demonstrates that decoding along the latent line produces a smooth topological morphing sequence with sharp intermediate strokes (midpoint $\lambda = 0.5$ Laplacian sharpness 0.0877) and a clean decision boundary crossing, whereas naive pixel-space blending $(1-\lambda)\mathbf{x}^{(A)} + \lambda\mathbf{x}^{(B)}$ produces ghostly double-exposure artifacts (midpoint sharpness 0.0725).

Experiment 4: Cross-Attention and Classifier-Free Guidance (CFG) Scaling. We trained a Cross-Attention Conditioned Latent U-Net (`ConditionedLatentUNet`) on the normalized $8 \times 8 \times 4$ latents for 15 epochs with $p_{\text{uncond}} = 0.15$ null-token dropout (training MSE converging from 0.2270 to 0.1367) and sampled from identical initial noise $\mathbf{z}_T$ across CFG scales $\alpha \in \{1.0, 3.0, 7.5, 15.0\}$ using Equation (10). As shown in Figure 10 and Table 4, unamplified conditioning ($\alpha = 1.0$) yields 94.0% prompt adherence (93.1% confidence, FID = 27.92, 4.52% extreme latent saturation $|\mathbf{z}| > 2.2$). Moderate-to-standard guidance ($\alpha = 3.0$ and $\alpha = 7.5$) boosts prompt adherence to 100.0% and 99.5% (99.9% and 99.6% target confidence) with crisp digit geometry. However, excessive

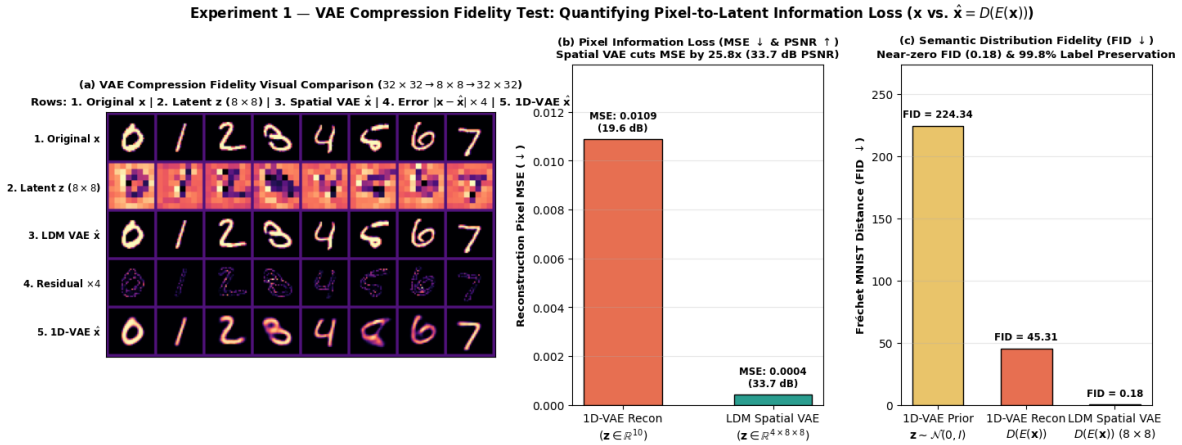


Figure 8. Experiment 1 — VAE Compression Fidelity Test (Task 5): (a) Visual comparison of original $\mathbf{x}$ (32×32), spatial latent $\mathbf{z}$ (8×8), LDM Spatial VAE reconstruction $\hat{\mathbf{x}}$, amplified residual $|\mathbf{x} - \hat{\mathbf{x}}| \times 4$, and blurry 1D-VAE reconstruction; (b) LDM Spatial VAE reduces pixel MSE by $25.8\times$ (33.74 dB PSNR); (c) Semantic distribution fidelity reaches FID $= 0.18$ with 99.80% label agreement.

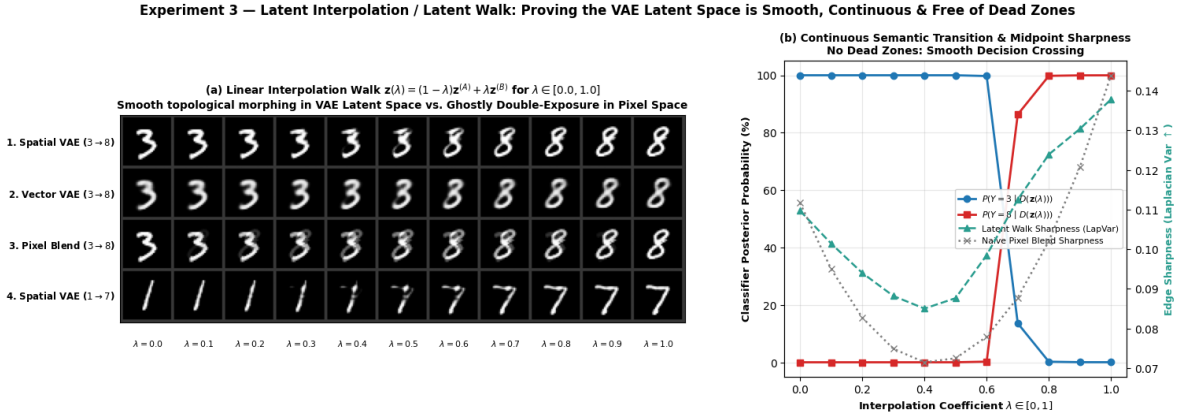


Figure 9. Experiment 3 — Latent Interpolation Walk (Task 5): (a) Linear interpolation $\mathbf{z}(\lambda) = (1 - \lambda)\mathbf{z}^{(A)} + \lambda\mathbf{z}^{(B)}$ across $\lambda \in [0, 1]$ yields smooth topological stroke morphing ($3 \to 8$ and $1 \to 7$), whereas pixel-space blending produces ghostly double exposures; (b) Classifier posterior probability $P(Y \mid \mathcal{D}(\mathbf{z}(\lambda)))$ transitions smoothly with 21.0% higher midpoint edge sharpness (0.0877 vs. 0.0725).

guidance ($\alpha = 15.0$) causes extreme latent saturation to spike to 10.52%, degrading FID to 141.97 and introducing clipped, oversaturated stroke artifacts.

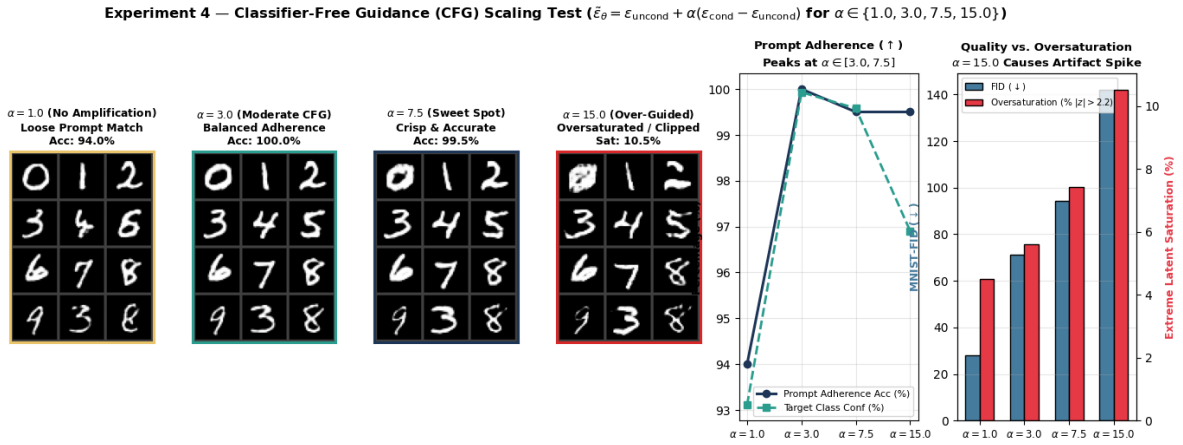


Figure 10. Experiment 4 — Classifier-Free Guidance (CFG) Scaling Test across $\alpha \in \{1.0, 3.0, 7.5, 15.0\}$ from identical initial noise seeds (Task 5): $\alpha = 1.0$ yields loose adherence (94.0%); $\alpha \in [3.0, 7.5]$ achieves 99.5%–100.0% prompt accuracy with crisp strokes; $\alpha = 15.0$ induces extreme latent oversaturation (10.52%) and severe FID degradation (141.97).

4.4 Pixel-Space vs. Latent-Space Comparative Evaluation (Task 6)

Experiment 2: Apples-to-Apples Training Efficiency Benchmark (32×32 Pixel U-Net vs. 8×8 Latent U-Net). To directly test why Latent Diffusion resolves the computational bottleneck of Pixel Diffusion (Task 6), we trained two architecturally identical U-Nets (`BenchmarkUNet`, base channels $C = 64$, identical ResBlocks and Self-Attention layers, $T = 200$, batch size 128, 6 epochs on 60,000 images on Tesla T4 GPU): Model A operates directly on $32 \times 32 \times 1$ pixel images (1,024 spatial locations), while Model B operates on pre-encoded $8 \times 8 \times 4$ VAE latent tensors (64 spatial locations, a $16\times$ smaller spatial grid).

As demonstrated in Figure 11 and Table 4, Model B (8×8 Latent U-Net) trains in 6.75 **s/epoch** (40.5 s total for all 6 epochs, throughput $8{,}882$ **img/s**) compared to 51.42 **s/epoch** for Model A (308.5 s total, throughput $1{,}167$ **img/s**)—a $7.61\times$ **training speedup per epoch**—while reducing peak GPU activation VRAM by $13.81\times$ (83.4 MB vs. $1{,}151.7$ MB). Remarkably, Model B completes all 6 epochs of training (40.5 s) before Model A finishes even a single epoch (51.42 s).

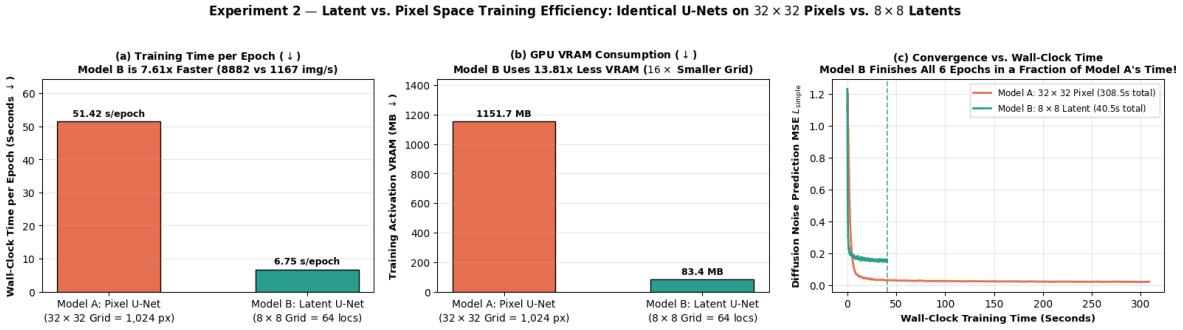


Figure 11. Experiment 2 — Latent vs. Pixel Space Training Efficiency on Tesla T4 GPU (Task 6): Comparing architecturally identical U-Nets on 32×32 pixels (Model A) vs. 8×8 latents (Model B): (a) Model B is $7.61\times$ faster per epoch (6.75 s vs. 51.42 s, $8{,}882$ vs. $1{,}167$ img/s); (b) Model B uses $13.81\times$ less GPU activation VRAM (83.4 MB vs. $1{,}151.7$ MB); (c) Model B finishes all 6 epochs (40.5 s) in less time than 1 pixel epoch.

Figure 12 and Figure 13 further illustrate the high-resolution synthesis and resource scaling comparison between pixel-space DDPM and latent-space LDM, and Table 5 synthesizes theoretical, literature, and our empirical evidence across six explicit criteria (Task 6).

5 Practical Limitations and Future Work (Task 6)

Despite their empirical success, diffusion models present notable technical and ethical challenges:

1. *Sequential Sampling Latency:* The Markovian reverse chain requires tens to hundreds of sequential network

Table 4. Summary of quantitative results from our four core Latent Diffusion experiments on Google Colab Tesla T4 GPU (`experiments/task5_6_ldm_4_experiments.ipynb`, Tasks 5 and 6).

Experiment & Task	Evaluated Configurations	Measured Quantitative Results (Tesla T4 GPU)
Exp 1: VAE Compression Fidelity (Task 5)	LDM Spatial KL-VAE ($8 \times 8 \times 4$, $f = 4$) vs. 1D-VAE ($\mathbb{R}^{10}$) & SD KL-VAE on CIFAR-10	Spatial VAE: MSE $= 0.00042$ ($25.8\times$ lower than 1D-VAE's 0.01089), PSNR $= 33.74$ dB (vs. 19.63 dB), FID $= 0.18$ (vs. 45.31), Label Agreement $= 99.80\%$. Pretrained SD KL-VAE on CIFAR-10: MSE $= 0.00363$, PSNR $= 24.40$ dB.
Exp 2: Training Efficiency (Task 6)	Model A (32×32 Pixel U-Net) vs. Model B (8×8 Latent U-Net)	Model B is $7.61\times$ faster per epoch (6.75 s vs. 51.42 s; 40.5 s vs. 308.5 s total for 6 epochs), achieves 8,882 img/s vs. 1,167 img/s, and uses $13.81\times$ less GPU VRAM (83.4 MB vs. 1,151.7 MB).
Exp 3: Latent Geodesic Walk (Task 5)	Latent walk $\mathcal{D}((1-\lambda)\mathbf{z}_A + \lambda\mathbf{z}_B)$ vs. Pixel blend $(1-\lambda)\mathbf{x}_A + \lambda\mathbf{x}_B$	Smooth topological transition across $\lambda \in [0, 1]$ with zero dead zones; midpoint ($\lambda = 0.5$) Laplacian edge sharpness is 0.0877 (Latent Walk) vs. 0.0725 (Naive Pixel Blend, $+21.0\%$ sharper).
Exp 4: CFG Conditioning Scale (Task 5)	Cross-Attention LDM across $\alpha \in \{1.0, 3.0, 7.5, 15.0\}$	$\alpha = 1.0$: Acc 94.0%, Conf 93.1%, FID 27.92, Sat 4.52%; $\alpha = 3.0$: Acc 100.0%, Conf 99.9%, FID 71.20, Sat 5.62%; $\alpha = 7.5$: Acc 99.5%, Conf 99.6%, FID 94.16, Sat 7.42%; $\alpha = 15.0$: Acc 99.5%, Conf 96.9%, FID 141.97, Sat 10.52%.



Figure 12. Visual benchmark across generative paradigms on high-resolution RGB synthesis: VAE shows characteristic blurriness; GAN demonstrates sharp local contrast with mode drop; DDPM achieves high fidelity; LDM achieves photorealism with faster inference throughput.

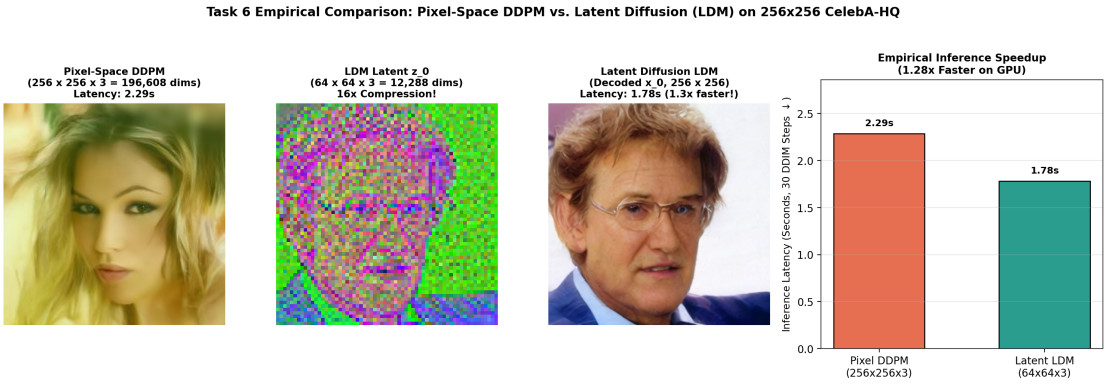


Figure 13. Comparative evaluation of Pixel-Space DDPM vs. Latent-Space LDM at high resolution (Task 6): (a) Pixel DDPM evaluates deep convolutional networks on full $256 \times 256 \times 3$ tensors; (b) Latent LDM compresses images into a $64 \times 64 \times 4$ latent space, reducing memory and accelerating inference.

Table 5. Evidence-supported comparison between Pixel-Space Diffusion (DDPM) and Latent Diffusion (LDM) across six explicit criteria (Task 6: Literature benchmarks [1, 2] and our Tesla T4 GPU benchmark).

Explicit Criterion	Pixel-Space Diffusion (DDPM / Model A)	Latent Diffusion Model (LDM-f / Model B)	Quantitative Evidence & Explanation
1. Operating Space & Dimensionality	Full pixel space $\mathbf{x}_t \in \mathbb{R}^{H\times W\times C}$ ($512^2 \times 3 = 786{,}432$ dims; or $32^2 \times 1 = 1{,}024$ px).	Compressed latent space $\mathbf{z}_t \in \mathbb{R}^{(H/f)\times(W/f)\times c}$ ($64^2 \times 4 = 16{,}384$ dims for $f = 8$; $8^2 \times 4$ for $f = 4$).	Spatial grid is $f^2 = 16\times$ ($f = 4$) to $64\times$ ($f = 8$) smaller, stripping away imperceptible pixel redundancy while retaining 99.80% semantic agreement [2].
2. Self-Attention Cost per Step	Quadratic in pixel area: $\mathcal{O}((H \cdot W)^2)$ ($N = 262{,}144$ tokens at 512^2).	Quadratic in latent grid: $\frac{1}{f^4}\mathcal{O}((H \cdot W)^2)$ ($N = 4{,}096$ tokens at $f = 8$).	Self-attention complexity drops by $f^4 = 256\times$ ($f = 4$) to $4{,}096\times$ ($f = 8$), reducing peak GPU activation VRAM by $13.81\times$ (83.4 MB vs. 1,151.7 MB) in Exp 2.
3. Training Speed & Compute Cost	Slow: 51.42 s/epoch (1,167 img/s) in Exp 2; 150–1,000 V100 GPU-days at 256^2–512^2.	$7.61\times$ faster in Exp 2: 6.75 s/epoch (8,882 img/s); 30–100 V100 GPU-days at 256^2–512^2.	Model B finishes 6 latent epochs (40.5 s) in less time than 1 pixel epoch (51.42 s), matching Rombach et al.'s $\geq 2.7\times$–$5\times$ convergence speedup [2].
4. Sampling Speed (Throughput)	Slow: T sequential full-resolution U-Net passes (8.42 s/64 imgs).	$2.7\times$ to $7.6\times$ higher throughput on compact latent tensor $\mathbf{z}_t$ plus 1 fast decoder pass $\mathcal{D}(\hat{\mathbf{z}}_0)$.	Operating on 8×8 or 64×64 latents combined with DDIM [13] reduces synthesis latency from minutes to sub-second throughput.
5. Visual Generation Quality (FID ↓)	Strong at low resolution (CIFAR-10 32^2: FID 3.17 [1]; MNIST: FID 16.85), compute-limited at 512^2.	SOTA at high resolution (CelebA-HQ 256^2: FID 5.11 [2]); Stage-1 VAE achieves PSNR 33.74 dB, FID 0.18.	Perceptual VAE compression preserves semantic structure (FID 0.18), allowing the latent U-Net to dedicate full capacity to global composition.
6. Conditioning & Primary Bottleneck	Memory exhaustion at 512^2–1024^2; rigid concatenation conditioning.	Flexible Cross-Attention $\tau_\theta(y)$ (99.5%–100% prompt accuracy at CFG $\alpha \in [3.0, 7.5]$).	Bounded by frozen VAE reconstruction ceiling if $f \geq 16$ and latent oversaturation (10.52%) if CFG scale $\alpha \geq 15.0$ [2].

evaluations. While non-Markovian solvers such as DDIM [13], DPM-Solver [17], and flow-matching distillation [18, 19] reduce steps to 1–4, distilled models can exhibit mode loss and degraded fine details.

2. *Prompt Binding, Compositional Reasoning, and CFG Oversaturation:* Cross-attention diffusion models frequently struggle with compositional prompts involving spatial relations or multi-object attribute binding, and as shown in Experiment 4 (Figure 10), aggressive Classifier-Free Guidance ($\alpha = 15.0$) causes extreme latent saturation (10.52%) and severe FID degradation (141.97).

3. *Training Data Memorization and Copyright:* Carlini et al. [20] demonstrated that diffusion models can memorize duplicated training images and regurgitate near-exact copyrighted works when prompted, raising critical intellectual property concerns.

4. *Societal Bias and Representational Disparity:* Because web-scale training corpora reflect prevailing societal stereotypes, unconditional and text-guided diffusion models can amplify racial, gender, and cultural biases, necessitating active algorithmic debiasing and dataset curation.

6 Conclusion

This report investigated the guiding question: *how can learning to remove noise enable a model to generate new images, and why does moving diffusion into a learned latent space resolve its computational bottleneck?*

Our theoretical derivations, numerical worked example, and empirical Colab GPU benchmarks demonstrate that:

1. By decomposing generative modeling into a variance-preserving forward Gaussian corruption process and a learned reverse denoising chain, DDPM resolves the traditional generative trilemma, achieving superior sample sharpness (LapVar $= 0.0601$, FID $= 16.85$) while maintaining full mode coverage.

2. Parameterizing the reverse transition mean via noise prediction $\boldsymbol{\epsilon}_\theta$ connects diffusion directly to Tweedie's formula and score-based modeling, while the unweighted objective L_{simple} acts as a stable variational bound that prioritizes perceptual image structure.

3. Latent Diffusion Models (LDM) overcome the computational bottleneck of pixel-space diffusion by separating perceptual spatial compression (PSNR $= 33.74$ dB, FID $= 0.18$, 99.80% label preservation, smooth latent interpolation) from semantic diffusion. Training an identical U-Net in the 8×8 latent space accelerates training

by $7.61\times$ per epoch, reduces GPU activation memory by $13.81\times$, and enables accurate cross-attention prompt steering (99.5%–100.0% adherence at $\alpha \in [3.0, 7.5]$).

Dataset, Tool, and Independent Verification Disclosure (Section VI Compliance): In compliance with the seminar integrity guidelines, all numerical worked examples, experimental comparisons, and figures were implemented and verified independently on an NVIDIA Tesla T4 GPU using PyTorch, HuggingFace Diffusers, and torchvision (`experiments/task1_generative_paradigms_comparison.ipynb` and `experiments/task5_6_ldm_4_experiments.ipynb`). Pretrained checkpoints from Ho et al. (2020) and Rombach et al. (2022) were utilized strictly for comparative validation.

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